

overlap-calculator Case Studies

Index of all 15 worked examples

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1 overlap-calculator Case Studies

This directory contains 15 worked case studies, each isolating one **user-selectable feature** of **overlap-calculator**. Case studies 01–11 ship in three documentation formats — Markdown, LaTeX source, and a typeset PDF. Case studies 12–15 ship in Markdown and LaTeX source (PDFs are built separately).

Each case study is intentionally small (**2 samples × 2 light sources**) so a full run finishes in seconds and the repository stays lean. The case isolates one knob at a time: light source, plot DPI, broadening width, Beer–Lambert reference, wavelength grid, sheet selection, etc.

1.1 Layout shared by every case

```
case_studies/<NN>_<name>/
+-- input.json
+-- submit.py
+-- README.md
+-- README.tex
+-- README.pdf
+-- output/
  +-- cli/
  |   +-- tables/{results,results_timings,descriptor_summary}.*
  |   +-- plots/...
  +-- api/
      +-- analysis_outputs.zip
      +-- tables/...
      +-- plots/...
```

`input.json` is the 2-entry manifest that points at the bundled files under [../input/files/](#). `submit.py` drives both the CLI and the HTTP API in one run. The Markdown README is the narrative for the case; the LaTeX and PDF carry the same content in typeset form. The `output/` tree is produced once you run `submit.py`; case study 07 produces no `plots/` directory because it turns plotting off.

1.2 Case study index

If this is your first time using `overlap-calculator`, start with **Case Study 01** — its **§1 Prerequisites** is the install / environment / API-startup walkthrough that every other case study links to instead of repeating.

Case	Folder	Feature exercised	Inputs	LaTeX	PDF
01	01_theoretical_only	Canonical TD-DFT-only workflow (anchor)	2 .out	tex	pdf
02	02_experimental_only	Measured spectra only	2 series in 1 .xlsx	tex	pdf
03	03_mixed_theory_experiment	Theoretical + experimental in one manifest	2 .out + 2 series	tex	pdf
04	04_custom_light_source	--light-source-file / light_source_files	2 .out + 1 CSV	tex	pdf
05	05_default_light_source_selection	--default-light-sources subset	2 .out	tex	pdf
06	06_plot_dpi	--plot-dpi 600 for publication TIFFs	2 .out	tex	pdf
07	07_table_only	--no-plot-outputs / plot_outputs=false	2 .out	tex	pdf
08	08_broadening_sigma	--sigma-ev (0.30 → 0.20 eV)	2 .out	tex	pdf
09	09_beer_lambert_tuning	--concentration-m and --path-cm	2 .out	tex	pdf
10	10_wavelength_grid	--wl-min / --wl-max / --num-points	2 .out	tex	pdf
11	11_sheet_overrides	sheet_name (CLI) / sheet_overrides (API)	2 series in 1 .xlsx	tex	pdf
12	12_frequency_resolved_prefactor	--prefactor-mode frequency-resolved	2 .out	tex	—
13	13_marcus_hush_width	--sigma-mode marcus-hush + --reorganization-ev	2 .out	tex	—
14	14_calibration_block	--calibration PATH (TOML; CLI/library only)	2 .out	tex	—
15	15_vibronic_tabular_branch	Vibronic/Franck-Condon spectra via experimental branch	2 series in 1 .xlsx	tex	—

1.3 Bundled inputs

Every case study draws from the same files in `../input/files/`:

- 13 Gaussian TD-DFT outputs (the case studies use 2 of them to keep runs short).
- One PhotochemCAD-derived Excel workbook with 8 numeric absorbance series (the case studies use 2 of them).

Citation for the experimental workbook. The bundled experimental absorbance data is derived from M. Taniguchi & J. S. Lindsey, "Database of absorption and fluorescence spectra of >300 common compounds for use in PhotochemCAD", *Photochem. Photobiol.* **94** (2018) 290–327. See the project [README](#) for the full acknowledgement and BibTeX entries (covering both the bundled light sources and the experimental workbook).